

Figure 1.49

For the curve $f(x) = \sqrt{x}$, we have $f(-x) = \sqrt{-x}$ and hence $f(-x) = \sqrt{-x}$ where $x < 0$, is the reflection of $f(x) = \sqrt{x}$ about y-axis. (See Figure 1.49.)

Illustration 1.7 Consider the functions:

$$\text{(i) } y = e^x \quad \text{(ii) } y = \log_e x.$$

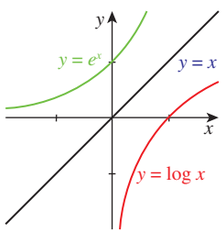


Figure 1.50

We know that, $y = e^x$ is the inverse function of $y = \log_e x$ and hence $y = e^x$ is the reflection of $y = \log_e x$ about $y = x$. (See Figure 1.50.)

Translation

A ***translation*** of a graph is a vertical or horizontal shift of the graph that produces congruent graphs.

The graph of	$y = f(x + c), c > 0$	causes the shift to the left.
	$y = f(x - c), c > 0$	causes the shift to the right.
	$y = f(x) + d, d > 0$	causes the shift to the upward.
	$y = f(x) - d, d > 0$	causes the shift to the downward.

Illustration 1.8 Consider the functions:

$$\text{(i) } f(x) = |x| \text{ (ii) } f(x) = |x - 1| \text{ (iii) } f(x) = |x + 1|$$

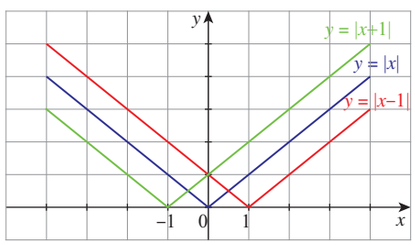


Figure 1.51